HW 3 — Escape from a Dielectric & Jackson 7.12

How to read this: black = write it down (sentences included — they are part of a good solution) · **red = the answer** · *teal = my note to you, do not copy* · grey box = optional

Problem 1 — Escape from a dielectric

> A point source of light is embedded near the flat surface of a dielectric of index n. Treat the emitted light as plane waves (rays) going out **isotropically**. Find the **fraction** of rays that can refract out into the vacuum above.

ANSWER

$$f = \frac{1}{2}\left[1 - \sqrt{1 - 1/n^2}\right]$$

Only rays inside a cone of half-angle i_0 about the normal get out, where $\sin i_0 = 1/n$.

*Glass ($n = 1.5$) gives **12.7%**; for large n it goes as **$1/4n^2$**.*

Step 1 — Find which rays can escape

Label each ray by its angle θ from the normal — that is its angle of incidence. **From Snell's law:**

$$n \sin\theta = n' \sin r$$ Lec 4, slide 7

Above is vacuum, so $n' = 1$:

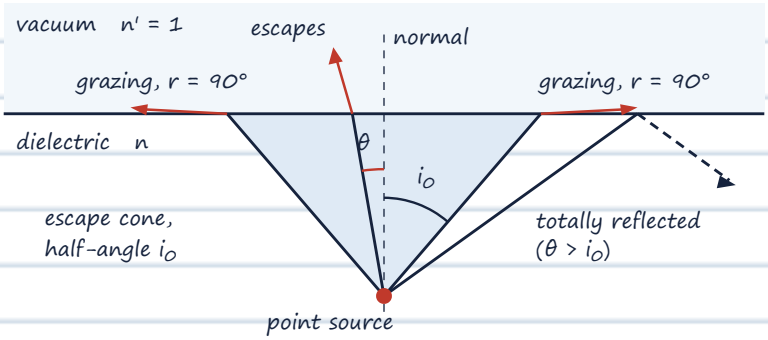
$$\sin r = n \sin\theta$$

But $\sin r \le 1$ always, **so** a refracted ray exists only if

$$n \sin\theta \le 1, \quad \text{that is} \quad \theta < i_0, \quad \text{where} \quad \sin i_0 = 1/n \quad (i_0 = \text{the critical angle})$$

If $\theta > i_0$: $\sin r > 1$, $\cos r$ imaginary → evanescent, **totally reflected**. Lec 4, slide 8

So the escaping rays fill a **cone of half-angle i_0** about the upward normal. (Downward rays never reach the surface.)



θ = a ray's angle from the normal; i_o = the cone's half-angle (the edge ray). ***Grazing:*** *at θ = i_o, sin r = n sin i_o = 1, so r = 90° — the ray comes out along the surface.*

Step 2 — Measure the size of that cone

Isotropic = equal power per solid angle, so fraction of rays = fraction of solid angle. **From spherical coordinates:**

$$d\Omega = \sin\theta \, d\theta \, d\phi$$

Integrate over the cone (θ: $0 \to i_o$, ϕ: $0 \to 2\pi$):

$$\Omega_{escape} = \int_0^{2\pi} d\phi \int_0^{i_o} \sin\theta \, d\theta = 2\pi \left[-\cos\theta\right]_0^{i_o} = 2\pi(1 - \cos i_o)$$

Step 3 — Turn the solid angle into a fraction

Whole sphere = 4π, so

$$f = \frac{\Omega_{escape}}{4\pi} = \frac{2\pi(1 - \cos i_o)}{4\pi} = \frac{1 - \cos i_o}{2}$$

Step 4 — Write the answer in terms of n

From Step 1 $\sin i_o = 1/n$, **and** $\sin^2 + \cos^2 = 1$:

$$\cos i_o = \sqrt{1 - \sin^2 i_o} = \sqrt{1 - 1/n^2}$$

Substitute into Step 3:

FINAL ANSWER

$$f = \frac{1}{2}\left[\, 1 - \sqrt{1 - 1/n^2} \,\right]$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

Problem 2 — Jackson 7.12

> The time dependence of electrical disturbances in good conductors is governed by the frequency-dependent conductivity (7.58). Consider **longitudinal** electric fields in a conductor, using Ohm's law, the continuity equation, and the differential form of Coulomb's law.
> **(a)** Show that the time-Fourier-transformed charge density satisfies $[\sigma(\omega) - i\omega\varepsilon_0]\rho(\mathbf{x},\omega) = 0$.
> **(b)** Using $\sigma(\omega) = \sigma_0/(1 - i\omega\tau)$ with $\sigma_0 = \varepsilon_0\omega_p^2\tau$, show that for $\omega_p\tau \gg 1$ any initial disturbance oscillates at the plasma frequency and decays with constant $\lambda = 1/2\tau$.

ANSWER

(a) $[\sigma(\omega) - i\omega\varepsilon_0]\,\rho(\mathbf{x},\omega) = 0$

(b) $\omega^2 + i\omega/\tau - \omega_p^2 = 0 \;\Rightarrow\; \omega \approx \pm\omega_p - \dfrac{i}{2\tau} \;\Rightarrow\; \rho \propto e^{\mp i\omega_p t}\, e^{-t/2\tau}$

— oscillates at ω_p, decays with $\lambda = 1/2\tau$.

Step 0 — The laws we are going to use

The problem names the first three. **Ohm's law:**

$$\vec{J} = \sigma(\omega)\,\vec{E}$$

Continuity (charge conservation):

$$\nabla\cdot\vec{J} + \partial\rho/\partial t = 0$$

Coulomb's law, differential form:

$$\nabla\cdot\vec{E} = \rho/\varepsilon_0$$

Plasma frequency (Lecture 6):

$$\omega_p^2 = \frac{Ne^2}{\varepsilon_0 m}$$ Lec 6, high-frequency behavior

Notation note. *The professor writes $\sigma_0 = Ne^2/(m\gamma_0)$; Jackson writes $\sigma_0 = \varepsilon_0\omega_p^2\tau = Ne^2\tau/m$. They are the same quantity, so $\tau = 1/\gamma_0$.*

Step 1 — Go into frequency space

Write $f(t) = f(\omega)\, e^{-i\omega t}$ (the usual convention). **Then** $\partial/\partial t \to -i\omega$, and **continuity becomes algebraic**:

$$\nabla\cdot\vec{J}(\mathbf{x},\omega) - i\omega\rho(\mathbf{x},\omega) = 0$$

Step 2 — Substitute Ohm's law, then Coulomb's law → part (a)

We want ρ alone. **Sub in** $\vec{J} = \sigma(\omega)\vec{E}$:

$$\sigma(\omega)\,\nabla\cdot\vec{E} - i\omega\rho = 0$$

Sub in $\nabla\cdot\vec{E} = \rho/\varepsilon_0$:

$$\frac{\sigma(\omega)}{\varepsilon_0}\,\rho - i\omega\rho = 0$$

×ε_0, factor out ρ:

FINAL ANSWER — PART (a)

$$\left[\,\sigma(\omega) - i\omega\varepsilon_0\,\right]\rho(\mathbf{x},\omega) = 0$$

Divide this by $(-i\omega)$ and it reads $[\varepsilon_0 + i\sigma(\omega)/\omega]\rho = 0$, i.e. $\varepsilon_{\text{eff}}(\omega)\,\rho = 0$ — the conductor permittivity $\varepsilon(\omega) = \varepsilon_r + ib/\omega$ from Lec 7, slide 2 *set to zero. A charge wave can only live where the permittivity vanishes.*

Step 3 — Put in the given $\sigma(\omega)$

A real disturbance has $\rho \neq 0$, so the bracket must vanish:

$$\sigma(\omega) = i\omega\varepsilon_0$$

Sub in the given $\sigma(\omega) = \sigma_0/(1-i\omega\tau)$, clear the fraction:

$$\frac{\sigma_0}{1 - i\omega\tau} = i\omega\varepsilon_0 \quad \Rightarrow \quad \sigma_0 = i\omega\varepsilon_0(1 - i\omega\tau) = i\omega\varepsilon_0 + \omega^2\varepsilon_0\tau$$

Sub in the given $\sigma_0 = \varepsilon_0\omega_p{}^2\tau$:

$$\varepsilon_0\omega_p{}^2\tau = i\omega\varepsilon_0 + \omega^2\varepsilon_0\tau$$

$\div\ \varepsilon_0\tau$ and rearrange:

$$\boxed{\omega^2 + \frac{i\omega}{\tau} - \omega_p{}^2 = 0}$$

Step 4 — Solve that quadratic for ω

Quadratic formula, $a = 1$, $b = i/\tau$, $c = -\omega_p{}^2$ (note $(i/\tau)^2 = -1/\tau^2$):

$$\omega = -\frac{i}{2\tau} \pm \sqrt{\omega_p{}^2 - 1/4\tau^2}$$

Step 5 — Apply the condition $\omega_p\tau \gg 1$

$\omega_p\tau \gg 1 \Rightarrow 1/4\tau^2$ negligible $\Rightarrow$ the root is just ω_p:

$$\omega \approx \pm\omega_p - \frac{i}{2\tau}$$

$\rho(t) \propto e^{-i\omega t}$, so

$$\rho(t) \propto e^{\mp i\omega_p t} \cdot e^{-t/2\tau}$$

Real part → oscillation, imaginary part → decay.

FINAL ANSWER — PART (b)

$$\text{oscillation at } \omega_p\,, \quad \text{decay constant } \lambda = \frac{1}{2\tau}$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

EXTRA — none of this page is required for the answer

OPTIONAL **Problem 1 — checks and numbers.** $n \to 1$ gives $f \to ½$ ✓ large n gives $f \approx 1/4n^2$ ✓ water 17.0% | glass 12.7% | diamond 4.5% | GaAs 2.1% — which is why LEDs need domes.

Jackson's footnote: why you must not use $\sigma = \sigma_0$

Keeping the DC value $\sigma \approx \sigma_0$ in part (a) would give

$$[\sigma_0 - i\omega\varepsilon_0]\rho = 0 \;\Rightarrow\; \omega = -i\,\sigma_0/\varepsilon_0 \;\Rightarrow\; \rho \propto e^{-(\sigma_0/\varepsilon_0)t}$$

ω is now purely imaginary, so there is **no oscillation at all**, and the decay constant is $\lambda_w = \sigma_0/\varepsilon_0 = \omega_p^2\tau$. For $\omega_p\tau = 100$ that is 20000× too fast — Jackson's "extremely rapid damping with the (wrong) decay constant". The reason is that the disturbance happens at $\omega \approx \omega_p$, where $\omega\tau \gg 1$, so the DC value of σ simply does not apply.

OPTIONAL **The decay constant is exact.** The imaginary part of both roots is exactly $-1/2\tau$ whenever $\omega_p\tau > ½$; only the oscillation frequency is approximated. For $\omega_p\tau = 100$ the exact roots are $\pm 99.9987/\tau - 0.5i/\tau$, against $\pm 100/\tau - 0.5i/\tau$.

OPTIONAL **Where ω_p came from.** Lecture 6 derives $\omega_p^2 = NZe^2/\varepsilon_0 m$ from the bound-electron model at high frequency, and quotes 60 MHz for the ionosphere and 6×10^{10}–6×10^{12} Hz for lab plasmas.